THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 01

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** Quadrant III	signs of (x, y) are $(-, -)$
2	**(c)** 13	$3(4) + (1) = 13$
3	**(a)** parallel	equal slopes, different intercepts
4	**(d)** $\frac{3}{5}$	distance from 0
5	**(a)** $9s^2 - 25t^2$	$(a+b)(a-b) = a^2 - b^2$
6	**(b)** infinitely many	centre can be anywhere on the perpendicular bisector
7	**(a)** 616 cm^2	$\pi r^2 = \frac{22}{7} \times 14^2 = 616$
8	**(c)** 4	multiply the number of outcomes of each experiment
9	**(d)** an AP with $d = 3$	constant difference 3
10	**(c)** 98	$a + (n-1)d = 2 + 24 \times (4) = 98$
11	**(a)** false; 8 is a multiple of 4 but not of 12	8 = 4×2 is a counterexample
12	**(a)** 9 factors; it is a perfect square	count the factor pairs; an odd count signals a repeated pair
13	**(d)** $\frac{w_1x_1 + w_2x_2 + \cdots + w_nx_n}{w_1 + w_2 + \cdots + w_n}$	each value counts by its weight
14	**(c)** 100	$\frac{100 \cdot 60 + 40y}{100 + y} = 50 \Rightarrow y = 100$
15	**(d)** data structure	data structures organise information
16	**(d)** 25	adjacent angles add to 180°: 6x + 30 = 180 ⇒ x = 25
17	**(c)** 3	rise ÷ run
18	**(a)** (2, 3)	substitute each point in the equation
19	**(a)** 190 cm^2	$2(lw + wh + lh) = 2(63 + 14 + 18) = 190$
20	**(c)** A is true but R is false	the cylinder has height $2r$; A is false

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$AB = \sqrt{4+4} = 2\sqrt{2}$, $BC = 2\sqrt{2}$, $AC = 4$. $AB = BC$, and $AB^2 + BC^2 = 8 + 8 = 16 = AC^2$: a right-angled isosceles triangle.	1+1
22	$s = 6$; area $= \sqrt{6 \times 3 \times 2 \times 1} = 6$ cm^2. Right-angled: $\frac{1}{2} \times 3 \times 4 = 6$ ✓	1+1
23	$\frac{112}{200} = 0.56$. The theoretical value is 0.5; with more tosses the experimental value tends to get closer.	1+1
24	$t_n = 11 + (n-1)(-3) = 14 - 3n$ (1). Recursive: $t_1 = 11$, $t_n = t_{n-1} - 3$ (1).	1+1
25	No (1). The chart gives only percentages of each shop's own total; Q's total could be much larger (1).	1+1
26	$\gcd(m, n) = \gcd(n, m-n)$ (1); every common divisor of m and n divides $m - n$ and conversely (1).	1+1
27	By the converse of the Midpoint Theorem Q is the midpoint of AC (1): $AQ = QC = 9$ cm (1).	1+1
28	Adding: $2x = 16$, so $x = 8$ (1). Then $y = 12 - 8 = 4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	$\frac{18}{30} = 0.6$ (1+1). Expected $= 0.6 \times 50 = 30$ (1).	2+1
30	$DE = \frac{1}{2}CA = 5$ cm, $EF = \frac{1}{2}AB = 7$ cm, $FD = \frac{1}{2}BC = 9$ cm (2). Perimeter $= 5 + 7 + 9 = 21$ cm (1).	2+1
31	$\frac{a+b}{2} - a = \frac{b-a}{2} > 0$ and $b - \frac{a+b}{2} = \frac{b-a}{2} > 0$. So $a < \frac{a+b}{2} < b$; it is rational as sums and quotients of rationals are rational.	1+1+1
32	Total $= 1000$ MW (1). Shares: coal 60%, solar 10%, wind 20%, hydro 10% (2).	1+2
33	For $x = 0, 1, 2$: $(0, -2)$, $(1, 1)$, $(2, 4)$. It cuts the y-axis at $(0, -2)$.	2+1
34	$2\pi r = 66 \Rightarrow r = 10.5$ m (1). CSA $= 2\pi r^2 = 2 \times \frac{22}{7} \times 110.25 = 693$ m^2 (1). Cost $= 693 \times 40 =$ ₹27,720 (1).	1+1+1

35	At 6 cm: $2\sqrt{100-36}=16$ cm; at 8 cm: $2\sqrt{100-64}=12$ cm (2). $16>12$ and $6<8$: the chord nearer the centre is longer (1).	2+1
	OR: $4k+5k=180^\circ \Rightarrow k=20^\circ$ (2); $\angle A=80^\circ, \angle C=100^\circ$ (1).	2+1
36	(i) $(7,3)\to(3,4)\to(4,3)\to(3,1)\to(1,2)\to(2,1)\to(1,1)\to(1,0)$ i.e. 5 applications of Step 3 (1). (ii) $(7,3)\to(3,1)\to(1,0)$: 2 reductions (1). (iii) Division is shorter by 3 reductions (1).	1+1+1
	OR: $825=375\times2+75$; $375=75\times5+0$ (2). gcd $=75$, in 2 reductions (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Stacked: totals and absolute components can be compared (but later segments do not start at 0); 100% stacked: relative shares can be compared but actual numbers cannot (2½). (ii) Stacked: which family spent the most overall? 100% stacked: which family has the smallest share for healthcare? (2½)	2½+2½
	OR: (i) Salt $=30$ mL in 900 mL ⇒ 3.33% (2). (ii) Sugar $=24$ mL in 900 mL ⇒ 2.67% (2). (iii) The other solution dilutes each substance as the total volume increases (1).	2+2+1
38	Let M be the midpoint of AD. In $\triangle ADC$: $MG \parallel DC$, $MG=\frac{1}{2}DC=10$ cm; in $\triangle ABD$: $MH \parallel AB$, $MH=\frac{1}{2}AB=6$ cm (2). As $AB \parallel DC$, both MG and MH are the line through M parallel to AB, so M, G, H are collinear (2). $GH=MG-MH=4$ cm and $GH \parallel AB$ (1).	2+2+1
	OR: Rotate a copy of $\triangle ABC$ by 180° about the midpoint of BC: the two triangles form a parallelogram (2). Its opposite sides are equal and parallel — it is the same parallelogram tile for each pair (1). Parallelograms tile the plane by repeating the grid of two sets of parallel lines (the sticks picture) (2).	2+1+2
39	$x=0.\overline{142857}$, $10^6x=142857.\overline{142857}$, $999999x=142857 \Rightarrow x=\frac{142857}{999999}=\frac{1}{7}$ (3). $2\times142857=285714$, $3\times142857=428571$ — same digits cycled (2).	3+2
	OR: (i) -12 and 12. (ii) $-850+1200-450=-100$ ⇒ debt of ₹100. (iii) Removing 4 debts of ₹3 each makes one ₹12 richer.	2+2+1
40	(i) $b=5$; $13=4a+5 \Rightarrow a=2$. (ii) $y=2x+5$. (iii) $2(10)+5=25$ ✓ yes. (iv) $y=0 \Rightarrow x=-2.5$: $(-2.5,0)$.	2+1+1+1
	OR: Points: $(0,0),(1,3)$ and $(0,0),(1,-3)$. (i) At the origin. (ii) Slopes 3 and -3; same steepness, opposite directions. (iii) Positive slope ⇒ growth; negative slope ⇒ decay.	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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